

The empirical recurrence for OEIS A232152: recovering a conjecture that is not written down

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Abstract

OEIS A232152 records “Empirical recurrence of order 58”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 153$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 58, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 58 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A232152 is “Number of $4 \times n$ 0..2 arrays with every 0 next to a 1 and every 1 next to a 2 horizontally, diagonally or antidiagonally.”. It has offset 1 and begins

1, 887, 67311, 3409361, 191897041, 11346946019, 663248443295, 38578051319991, ...

The entry states:

Empirical recurrence of order 58 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:55:30, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 6643$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 153$ of the 6643, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 153$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 306$ exact terms of the model returns order 58 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{58} c_i a(n-i),$$

c_1	40	c_{16}	45836010162865	c_{31}	−37670843524284455309	c_{46}	137301679336942669
c_2	730	c_{17}	1663280456621585	c_{32}	−41013323038790286360	c_{47}	−9258271215451758
c_3	14953	c_{18}	1018349296027671	c_{33}	−58015215248853054676	c_{48}	−9449178220634194
c_4	270835	c_{19}	14438947601890913	c_{34}	−123363995333516481298	c_{49}	25274940369381577
c_5	38982	c_{20}	4609480027260821	c_{35}	28469020438775630696	c_{50}	−2499056746491598
c_6	−38277925	c_{21}	−109182204280070998	c_{36}	79989014300264886271	c_{51}	9041764231364382
c_7	−269432018	c_{22}	−67568606502741311	c_{37}	53266732396773071566	c_{52}	11537355533887033
c_8	−139645610	c_{23}	−905985122850630921	c_{38}	306728308872218378316	c_{53}	−2981876863451627
c_9	6476817876	c_{24}	−595027268142389834	c_{39}	−67144404922808597306	c_{54}	−1732965238287728
c_{10}	81078145505	c_{25}	215254073468827810	c_{40}	−42999334796544917916	c_{55}	3352210491350384
c_{11}	304386087096	c_{26}	383538183317417156	c_{41}	−31218100190642939364	c_{56}	−291236340926251
c_{12}	−1366264774725	c_{27}	9849214239862169651	c_{42}	−334744517809799871120	c_{57}	−152565623795220
c_{13}	−5541922345789	c_{28}	13951186392546057549	c_{43}	117757680594247844928	c_{58}	3981150301401907
c_{14}	−3090982470889	c_{29}	8585137027511333040	c_{44}	67816820666185209936		
c_{15}	−117914785619090	c_{30}	27275558465446917425	c_{45}	39592905690559861728		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 58, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $58 + 4$ terms removed returns order 58 as well, so $\deg q_{\min} = 58 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $58 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 58 that this sequence satisfies — and that family turns out to have exactly one member.

References

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